

DS	Session	Stack	Question Information	Level 1	Challenge 45
----	---------	-------	----------------------	---------	--------------

Problem Description:

Sajid was booking a tour package of IRCTC from Chennai to Delhi for his family.

Two of the relatives was interested in joining to this tour.

these two persons are studying engineering in computing technology. only one tickets are remaining in the IRCTC portal.

So, Sajid decided to book one ticket for out of those persons also along with his family members.

He wants to identify the one person out of these persons. he decided to conduct a technical task to identify the right person to travel.

the task was that, Check for Balanced Brackets in an expression using Stack

Can you help them to complete the task?

Question No:5

Function Description

- Declare a character stack S.
- Now traverse the expression string exp.
 1. If the current character is a starting bracket ('(' or '{' or '[') then push it to stack.
 2. If the current character is a closing bracket (')' or '}' or ']') then pop from stack and if the popped character is the matching starting bracket then fine else brackets are not balanced.
- After complete traversal, if there is some starting bracket left in stack then "not balanced"

Input Format

Single line represents only braces (both curly and square)

Output Format

If the given input balanced then print as Balanced (or) Not Balanced

Logical Test Cases

Test Case 1

INPUT (STDIN)

EXPECTED OUTPUT

{ () } []

Balanced

Test Case 2

INPUT (STDIN)

EXPECTED OUTPUT

{ ({()}) } Not Balanced

Code:

```
1  #include <stdio.h>
2  #define MAX 100
3
4  char stack[MAX];
5  int top = -1;
6
7  void push(char ch)
8  {
9      if (top < MAX - 1)
10     {
11         top++;
12         stack[top] = ch;
13     }
14 }
15
16 char pop()
17 {
18     if (top == -1)
19         return '\0';
20
21     return stack[top--];
22 }
23
24 int isMatching(char open, char close)
25 {
26     if (open == '(' && close == ')')
27         return 1;
28
29     if (open == '{' && close == '}')
30         return 1;
31
32     if (open == '[' && close == ']')
33         return 1;
34
35     return 0;
36 }
```

```

37
38 int main()
39 {
40     char str[MAX];
41     int i;
42     int balanced = 1;
43
44     scanf("%s", str);
45
46     for (i = 0; str[i] != '\0'; i++)
47     {
48         /* Opening brackets */
49         if (str[i] == '(' ||
50             str[i] == '{' ||
51             str[i] == '[')
52         {
53             push(str[i]);
54         }
55
56         /* Closing brackets */
57         else if (str[i] == ')' ||
58                 str[i] == '}' ||
59                 str[i] == ']')
60         {
61             char open;
62
63             if (top == -1)
64             {
65                 balanced = 0;
66                 break;
67             }
68
69             open = pop();
70

```

```
71         if (!isMatching(open, str[i]))
72     {
73         balanced = 0;
74         break;
75     }
76 }
77 }
78
79 /* Stack must be empty at the end */
80 if (top != -1)
81 {
82     balanced = 0;
83 }
84
85 if (balanced)
86     printf("Balanced");
87 else
88     printf("Not Balanced");
89
90 return 0;
91 }
```

Output:
Test Case-1

```
{()}[]  
Balanced  
-----  
Process exited after 30.68 seconds with return value 0  
Press any key to continue . . . |
```

Test Case-2

```
{{({}})]}  
Not Balanced  
-----  
Process exited after 2.27 seconds with return value 0  
Press any key to continue . . . |
```

Keshav Kumar Rauniyar

CH.SC.U4CSE25236

